

Section A: mathematics

1. Selling Price of the bicycle

Given

$$CP : 2400$$

$$\text{Profit} : 15\%$$

$$\text{Profit} = 15\% \text{ of } 2400$$

$$\text{Profit} = \frac{15}{100} \times 2400 = 360$$

$$SP = CP + \text{Profit}$$

$$= 2400 + 360$$

$$= 2760$$

8.2. area of right angled triangle

given

$$\text{base} = 12\text{cm}$$

$$\text{height} = 9\text{cm}$$

Formula

$$A = \frac{1}{2} * b * h$$

$$A = \frac{1}{2} * 12 * 9$$

$$A = 54\text{cm}^2$$

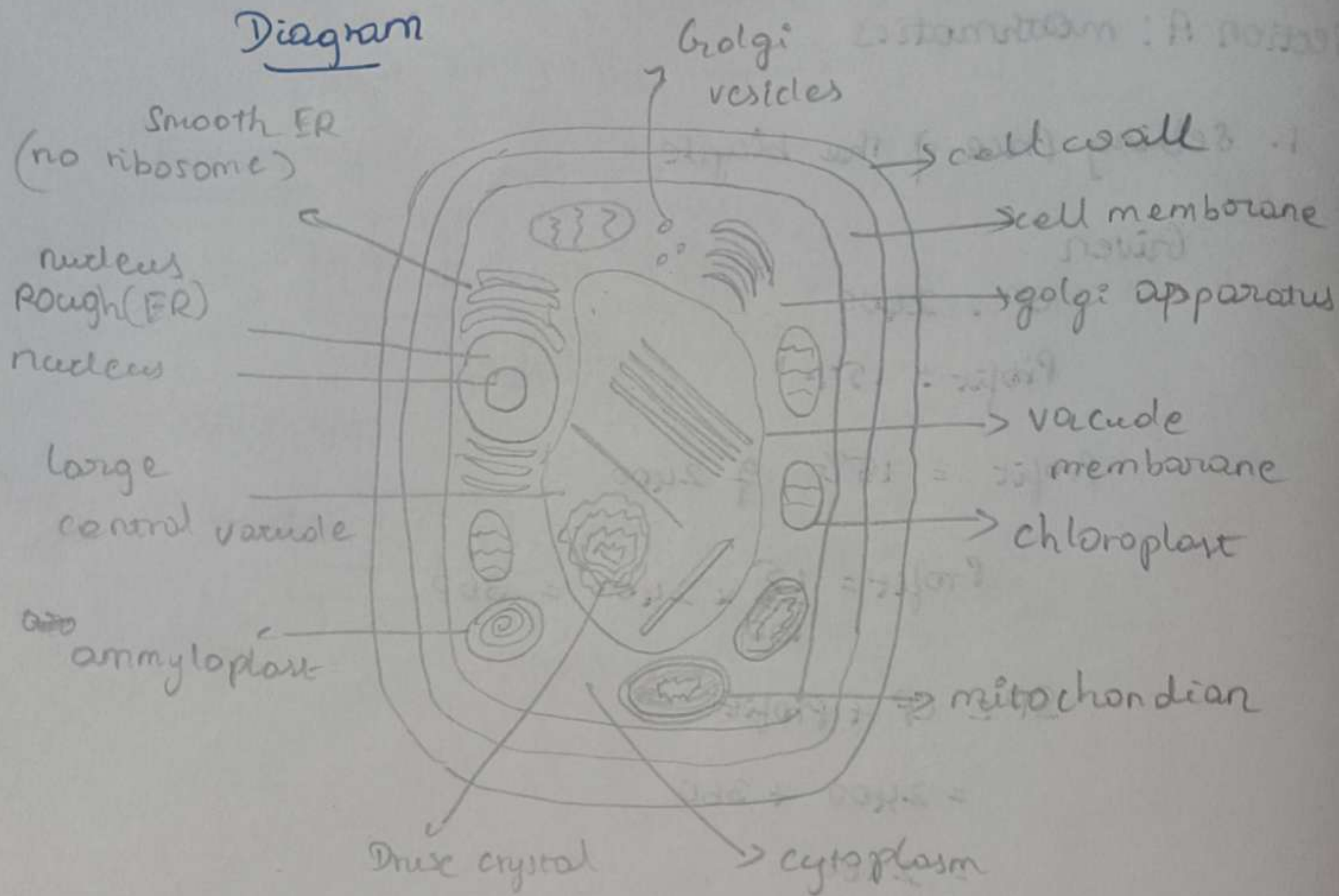
Section B Biology

2. plant cell

A plant cell contains a cell wall, cell membrane, cytoplasm, nucleus, chloroplasts and a large central vacuole.

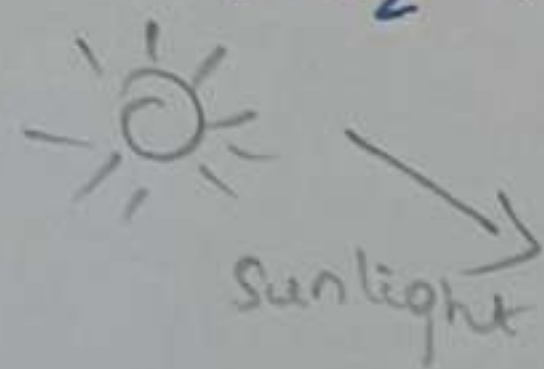
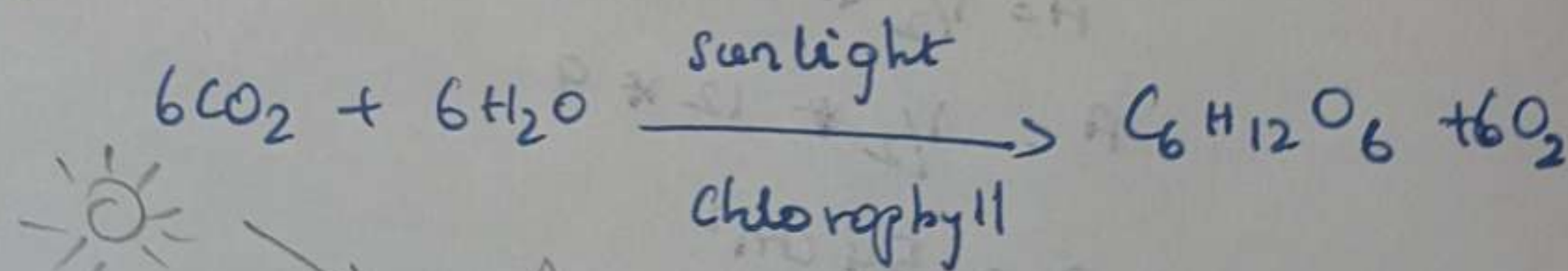
~~For the~~

Diagram

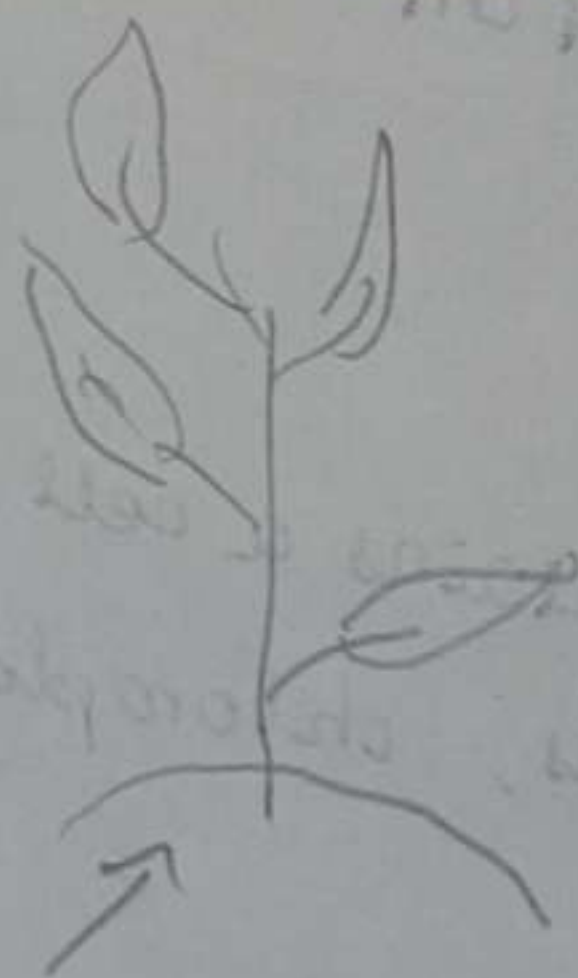


9. Photosynthesis

Photosynthesis is a process by which green plants prepare their food using CO_2 and H_2O in the presence of sunlight and chlorophyll, producing glucose and oxygen.



Sunlight



O_2

Glucose

water

CO_2

physics section : c

3 (a) Derivation of the first equation of motion

$$a = \frac{\text{change in velocity}}{\text{time}} = \frac{V - u}{t}$$

$$at = V - u$$

$$V = u + at$$

3 (b)

$$V = u + at$$

$$25 = 5 + 4a$$

$$20 = 4a$$

$$a = 5 \text{ m/s}^2$$

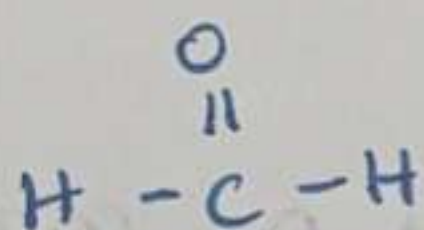
section D GK

4) Jaipur is the largest planet in solar system

10 DR. B.R. Ambedkar

section E Chemistry

5(a) methanol has the molecular formula HCHO



methanal contains the aldehyde functional group ($-\text{CHO}$)

answer:

compound : methanal

Formula : HCHO

functional group : Aldehyde ($-\text{CHO}$)

5(b) sodium - symbol - (Na) | group = 1
atomic number - 11 | period = 3

physical property = sodium is soft and silvery white material

Section F: Computer science

6. check number is prime or not

Python

```
num = int(input("Enter a number"))
```

```
if num <= 1:
```

```
    print("not a prime number") else: is_prime = True
```

```
for i in range(2, int(num**0.5) + 1):
```

```
    if num % i == 0:
```

```
        is_prime = False
```

```
        break
```

```
if is_prime:
```

```
    print("prime number")
```

```
else:
```

```
    print("not a prime number")
```

eg:

Enter a number: 17

Prime number

7(a) water cycle

The water cycle is a continuous movement of water between the earth and atmosphere. Water evaporates due to the sun's heat, forms clouds through condensation, and falls back as rain through precipitation.

~~7(a)~~ 7(b) Newton's Third law

It states that for every action, there is an equal and opposite reaction. For example when we push a wall, it pushes us back with an equal force in the opposite direction.